

BOOVANESH KUMAR

Backend Developer (Golang) — Microservices | Full-Stack (Vue.js) | Cloud & DevOps (AWS, Docker)

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PROFESSIONAL SUMMARY

Backend Developer (Golang) with around 3 years of experience architecting Go-based microservices, distributed systems, and REST APIs, complemented by full-stack development in Vue.js and cloud/DevOps exposure (AWS, Docker, Kubernetes). Skilled in performance tuning, scalability, load balancing, and event-driven architectures, with strong production debugging experience. Experienced in Agile Scrum, working in sprint-based development with story estimation, sprint planning, and cross-functional collaboration.

TECHNICAL SKILLS

Languages: Go (Golang), JavaScript, TypeScript

Backend & Architecture: Data Structures & Algorithms, Microservices, gRPC, Protocol Buffers, Kafka, REST APIs, Goroutines, Channels, Worker Pool / Fan-In / Fan-Out concurrency, Event-Driven Design, Distributed Systems, Scalability, Load Balancing, Horizontal Scaling, Monolith, Monorepo

Frontend: Vue.js, API Integration, Dashboards, ReactJS (fundamentals)

Databases: MongoDB, PostgreSQL, Cassandra, Redis

Cloud & DevOps: AWS (S3, Lambda, SQS, CloudWatch), Docker, Kubernetes, Jenkins, GitLab CI/CD, Linux, Shell Scripting, Terraform

Monitoring & Tools: Elasticsearch, Kibana, APM, pprof, AI-assisted development tools

PROFESSIONAL EXPERIENCE

Software Engineer — Mitrahsoft

- Built concurrent background-job pipelines (Worker Pool, Fan-In/Fan-Out) and resolved Kubernetes pod crashes via controlled goroutine execution.
- Developed microservices using gRPC and Kafka-based event-driven pipelines, with Elasticsearch/Kibana/APM monitoring and alerting.
- Developed automated test cases for desktop application testing using Playwright (TypeScript) and RobotJS.
- Implemented multilingual support using go-i18n and loaded application logs into Snowflake for analytical reporting.
- Developed a serverless Campaign Mail Scheduler (AWS Lambda, SQS, SendGrid) for event-driven bulk email processing.
- Served as an LLM Trainer (Turing) on Amazon Nova prompt engineering and response evaluation, achieving ~95% unit test coverage.
- Delivered cross-stack features spanning Vue.js dashboards, AWS S3 file handling, OAuth integrations, and CI/CD automation (Jenkins, Docker).

Front-End Development Trainee — Besant Technologies

- Completed a 6-month intensive ReactJS front-end training program, building practice applications covering component design, state management, and API integration.

KEY PROJECTS

Ecommerce Application

- Integrated Firebase Authentication with backend JWT verification (Firebase Admin SDK, Golang) and RBAC for USER/ADMIN roles; used AES encryption for sensitive data.
- Built a scalable Order Management and Inventory System with transactional consistency and idempotent APIs to prevent duplicate transactions.
- Designed a Wallet System using double-entry ledger for ACID-compliant transactions, with fraud detection, transaction validation, and audit logging.
- Integrated Stripe payments (sandbox), Firebase Cloud Messaging for real-time notifications, and Firestore for real-time order tracking/chat sync; built a WebSocket-based rule-driven chatbot in Golang for customer support workflows.

HRMS — Human Resource Management System

- Improved legacy code and MongoDB query performance by adding indexes and removing nested loops, cutting report generation time from 5 minutes to 25 seconds for 1 lakh (100,000+) records.
- Implemented Go concurrency patterns (Worker Pool, Fan-In/Fan-Out) for background PDF and Excel report generation.
- Resolved Kubernetes pod crashes from high-memory jobs using controlled goroutine execution and gopsutil-based memory checks, improving scalability under load.
- Integrated Elasticsearch/Kibana dashboards and alerting; built AWS S3 file upload, encryption, and pre-signed URL features.
- Developed an Expense microservice with approval workflows and Tesseract OCR-based receipt data extraction.

Freight Management System — Microservices

- Implemented gRPC-based inter-service communication and a Kafka-driven event system for real-time parcel tracking.
- Built event deduplication logic and used Cadence workflow orchestration for retrying and reprocessing failed events.
- Generated freight bills and shipping labels via ZPL, and implemented fuel-price and carrier-discount business rules.
- Used PostgreSQL for transactional data and Cassandra for high-volume tracking events; managed schema migrations.
- Applied dependency injection (samber/do) to reduce coupling; wrote unit/integration tests using a TDD approach.

EDUCATION

Bachelor of Engineering — Mechatronics, Kongu Engineering College

2022

HSC — St. Paul's Matriculation Hr. Sec. School

2018

SSLC — St. Paul's Matriculation Hr. Sec. School

2016

CERTIFICATIONS

- AWS Certified Solutions Architect – Associate (Udemy)
- Cyber Security From Beginner to Expert (Udemy)
- Software Architecture and Design of Modern Large Scale Systems (Udemy)